Arun Rawat

Sector 76, Noida – 201301

+91-9259209601 arun.rawat.sde@gmail.com linkedin.com/in/arun-rawat-sde github.com/arunrawat14

Professional Summary

Applied AI & Full-Stack Engineer with hands-on experience building production-grade AI systems across voice automation, food-tech, edtech, and conversational AI domains. Skilled in developing autonomous AI agents, multimodal LLM workflows, RAG pipelines, scalable REST APIs, and end-to-end full-stack platforms using FastAPI, Laravel, Node.js, React, Vue.js, OpenAI, Gemini, Docker, vector databases, and cloud infrastructure. Experienced in independently owning complete product lifecycles from architecture design and backend engineering to deployment, optimization, and production scaling.

Technical Skills

Languages: Python, JavaScript, SQL, C++

AI & LLM: OpenAI API (GPT-4o, GPT-5), Gemini, LangChain, LangGraph, RAG, Prompt Engineering, Function Calling, Vector Search, Pinecone, FAISS, OpenAI Embeddings, Retell AI, Agentic Workflows

Backend & APIs: FastAPI, Flask, Laravel, Node.js, Express.js, RESTful APIs, Webhooks, AsyncIO, Pydantic, Gunicorn, Uvicorn

Databases: PostgreSQL, MySQL, MongoDB, Pinecone, FAISS

Frontend: Vue.js, React.js, React Native, Tailwind CSS

DevOps & Cloud: Docker, GitHub Actions (CI/CD), DigitalOcean, AWS, Nginx, Linux, Prometheus, Loki, Firebase

Tools & Libraries: FFmpeg, PyMuPDF, Playwright, aiohttp, Twilio, Brevo, Postman, Jira, ClickUp

Education

Graphic Era Hill University **Oct 2021 – Jun 2025**

Bachelor of Science in Computer Science **Dehradun, Uttarakhand, India**

Experience

Software Engineer **Aug 2024 – Present**

PenIT Technology Pvt. Ltd. **Noida, India**

AI Voice Sales Agent **Mar 2026 – Apr 2026**

Retell AI · FastAPI · OpenAI · Function Calling · Webhooks · Docker

- Engineered and deployed a production AI outbound voice agent using Retell AI, handling **300+ calls/day** and fully automating CSCS card sales – lead qualification, product recommendation, order creation, and payment-link delivery.
- Delivered **8+ LLM function-calling integrations** covering real-time product lookup, order management, callback scheduling, VAT validation, and Klarna fallback payment handling – orchestrated entirely through the conversational AI.
- Crafted prompt-engineered conversational flows with customer qualification logic, multi-step objection handling, fixed pricing enforcement, and compliance guardrails to maintain consistent and controlled sales interactions.
- Implemented webhook-driven call lifecycle processing for recording ingestion, AI-generated summaries, call disposition tagging, and post-call analytics using async FastAPI handlers with Pydantic schema validation.

AI Sales Call Analysis Microservice **Feb 2026**

FastAPI · OpenAI Whisper · FFmpeg · AsyncIO · Docker · Webhooks

- Deployed an async FastAPI microservice for AI-powered sales call auditing – accepting audio uploads and executing configurable analysis workflows with runtime-selectable STT models, LLM models, and JSON report schemas.
- Preprocessed recordings via **FFmpeg** into mono 16 kHz WAV format, then chunked and concurrently transcribed large audio files using **OpenAI Whisper + async semaphores** to reduce end-to-end processing time.
- Integrated GPT-based analysis on transcripts to produce structured audit reports with scoring, checklist validation, issue detection, and actionable feedback; delivered results via **webhook callbacks** and background task execution.

AI Fertility Matching Agent (Expecting.AI) **Aug 2025 – Sep 2025**

LangGraph · OpenAI · Gemini · Pinecone · FAISS · FastAPI · Docker

- Established a reusable AI microservice platform with dynamic project-based routing, enabling multiple chatbot agents across different domains to operate on a single shared backend infrastructure.
- Launched a **production-grade RAG pipeline** using LangGraph, OpenAI embeddings, Pinecone, and FAISS supporting **25+ routing intents** – surrogate, egg donor, sperm donor, clinic, agency, insurance, escrow, therapist, and provider discovery.
- Developed NLP-based filter extraction for **10+ structured attributes** (age, BMI, location, ethnicity, blood group, insurance) and structured a multi-index vector search architecture across separate profile, provider, and knowledge-base indexes for accurate low-latency retrieval.
- Authored ingestion pipelines for **PDF, DOCX, and JSON** sources using PyMuPDF, python-docx, and LangChain text splitters with Pinecone batch upserts; implemented conversation memory and follow-up query chaining for contextual interactions.

Coteca LMS | Learning Management System **Apr 2025 – May 2025**

Vue.js · Laravel · MySQL · CRM APIs · Docker · ClickUp

- Led end-to-end technical planning and delivery of Coteca LMS – a web platform for **100+ monthly users** covering online course delivery and CITB certification test preparation.
- Produced **2 core modules**: a CITB Mock Exam Engine for timed UK certification practice, and an LMS module supporting modular content delivery across **5 layout types** with video, HTML, text, and image formats.
- Implemented Laravel + MySQL backend for course creation, module structuring, user assignment, progress tracking, and conditional test-access logic with feature-based access control tied to purchased services.
- Integrated the client's **CRM API** to sync learner activity, auto-flag exam-eligible users, and trigger lead generation for unpurchased areas; mentored 1 junior developer across the full project lifecycle.

Projects

Ingrify | AI Food Intelligence Platform (Backend) **Dec 2025 – Apr 2026**

FastAPI · MongoDB · GPT-4o/GPT-5 · Docker · DigitalOcean

- Spearheaded a **47+ endpoint** FastAPI + MongoDB backend powering an AI food intelligence platform – modules spanning authentication, AI scanning, product catalog, rewards, quizzes, notifications, and personalized dashboards.
- Shipped multimodal **GPT-4o and GPT-5** pipelines for ingredient OCR, barcode extraction, multi-image product validation, allergen detection, NOVA classification, and AI-generated health insights from raw food labels.
- Engineered user engagement systems including OTP auth, referral rewards, wallet and point-ledger management, leaderboard generation, badge unlocking, and personalized product recommendations driven by dietary profile data.
- Optimized high-traffic query performance via **MongoDB aggregation pipelines** ($lookup, $facet, $sample), TTL-based namespace caching, and MongoDB indexing; integrated Firebase push notifications, Twilio OTP, Brevo email, Nginx, Prometheus, Loki, and GitHub Actions CI/CD.
- Programmed automated catalog ingestion tooling using **Playwright + aiohttp + OpenAI Vision** to scrape, validate, and enrich **1,000+ food products**, eliminating manual data-entry workflows entirely.

EdTech Platform | *React, Tailwind CSS, Node.js, Express.js, MongoDB, Razorpay* **Jan 2024**

- Assembled a full-stack education marketplace enabling students to enroll in courses and instructors to publish, price, and manage content – integrated end-to-end with **Razorpay** for payment processing.
- Configured dynamic instructor dashboards for course content management, pricing control, and student engagement tracking.
- Created personalized user profiles for both learners and educators with a clean, responsive UI using Tailwind CSS.

Leadership & Extracurricular

Team Lead, Hackathon | *Blockchain-Based Counterfeit Detection App* **Apr 2023**

- Led a team to design and deliver a Web 3.0 counterfeit-product verification app; managed task ownership, architecture decisions, and on-time delivery under hackathon constraints.

Participant, Inter-College Hackathon | *AI Disease Prediction Web App* **Nov 2023**

- Coded a symptom-based ML disease prediction web app with a responsive UI, collaborating across a team to implement predictive models within a competitive timeline.